

Time Series Analysis 2026

Capstone Report | IIT Guwahati x StockGro

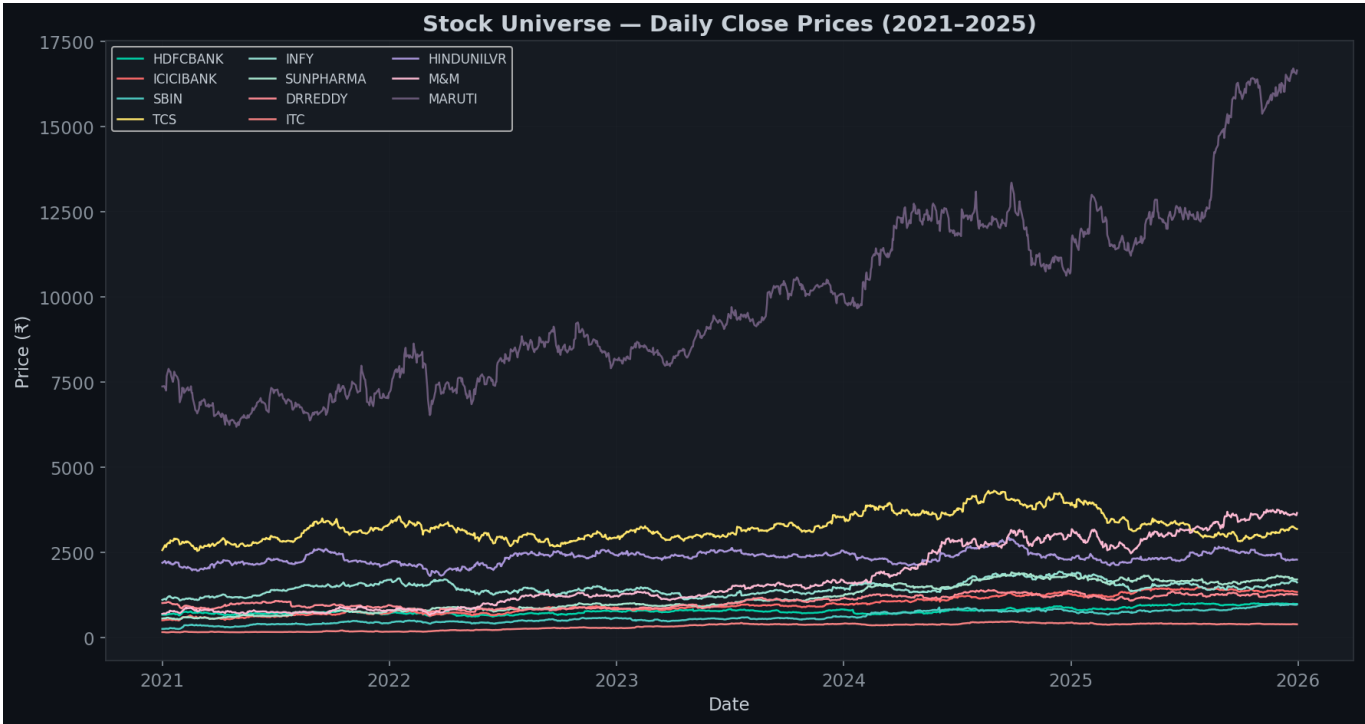
1. Methodology and Models Used

We applied time series forecasting to 11 NSE stocks, allocated Rs.10,00,000 in a virtual portfolio on StockGro, and compared predictions against actual market outcomes. Our models include ARIMA, Prophet, LSTM, Exponential Smoothing, and GARCH.

Dataset: Yahoo Finance (yfinance), Jan 2021 - Dec 2025, daily.

Split: Training (< Jul 2025) | Testing (Jul-Dec 2025, 125 observations).

Model	Library	Purpose
ARIMA/SARIMA	pmdarima	Price forecasting (auto order via AIC)
Facebook Prophet	prophet	Trend + seasonality with Indian holidays
LSTM (2-layer)	PyTorch	Non-linear deep learning sequence model
Exp. Smoothing	statsmodels	Holt-Winters triple exponential smoothing
GARCH(1,1)	arch	Conditional volatility estimation

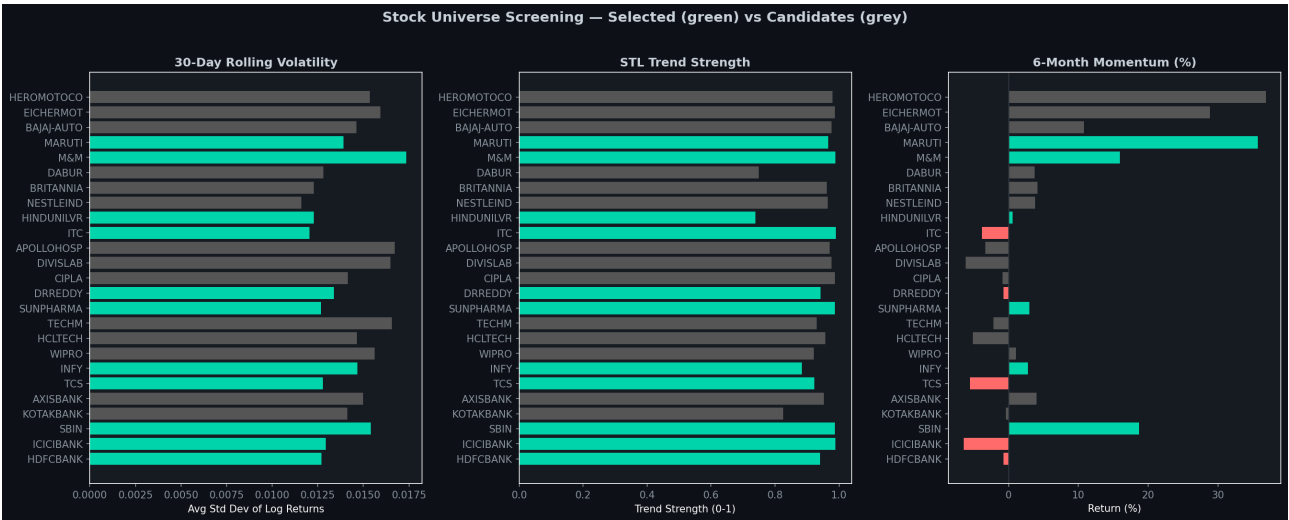


2. Stock Selection Rationale (Task 1)

We screened 25 candidate stocks across 5 sectors using three quantitative criteria: (1) 30-day rolling std dev, (2) STL trend strength, (3) 6-month sector momentum. Selected 11 stocks:

Sector	Selected Stocks	Key Justification
Banking	HDFC Bank, ICICI Bank, SBI	Highest liquidity (15-25M daily vol)
IT	TCS, Infosys	Top-2 by market cap, mean-reversion
Pharma	Sun Pharma, Dr. Reddy's	Low correlation, defensive sector
FMCG	ITC, Hindustan Unilever	Lowest volatility (4th-8th pctl)
Auto	M&M, Maruti	Highest sector momentum (+25.6%)

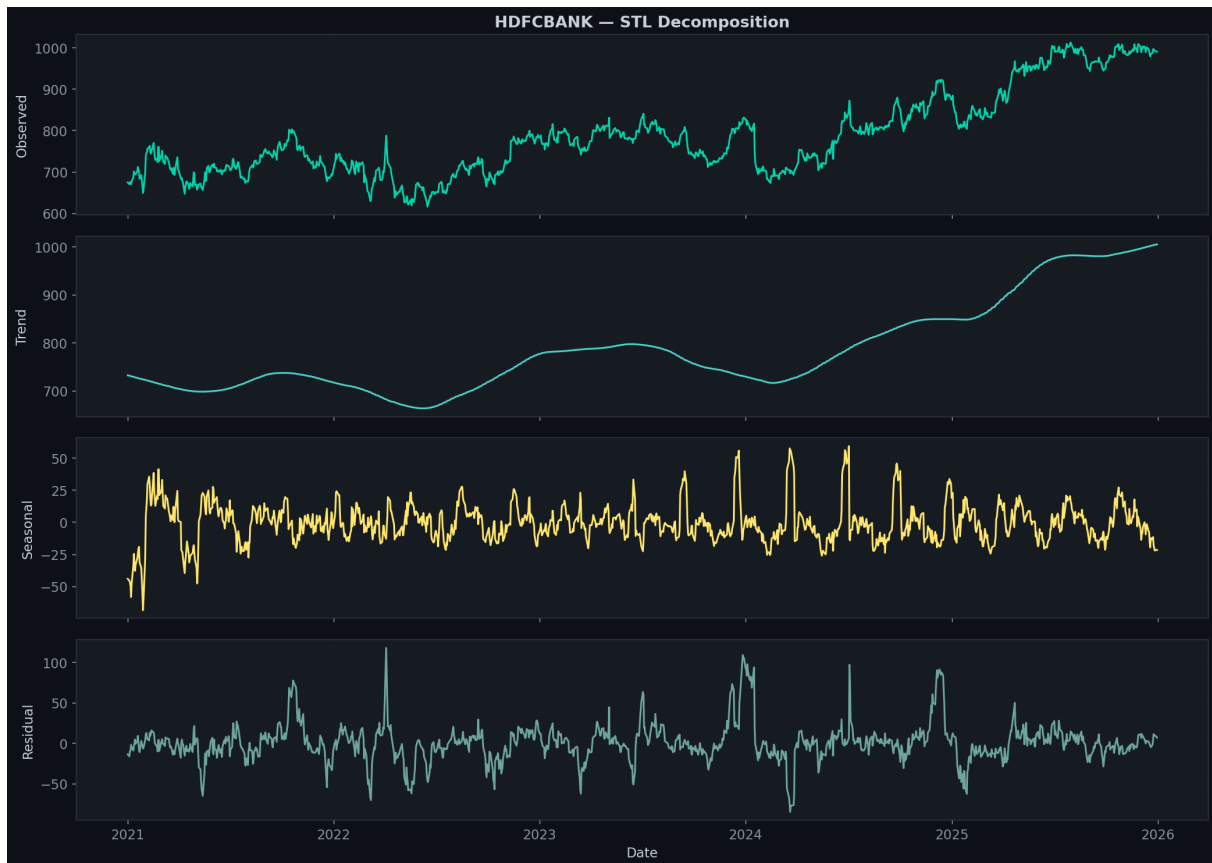
14 candidates were rejected due to lower liquidity, weaker trends, or redundancy within their respective sectors.



3. Data Preprocessing Steps (Task 2)

- Missing values: Forward-fill for market holidays and trading halts.

- Stationarity
differencing



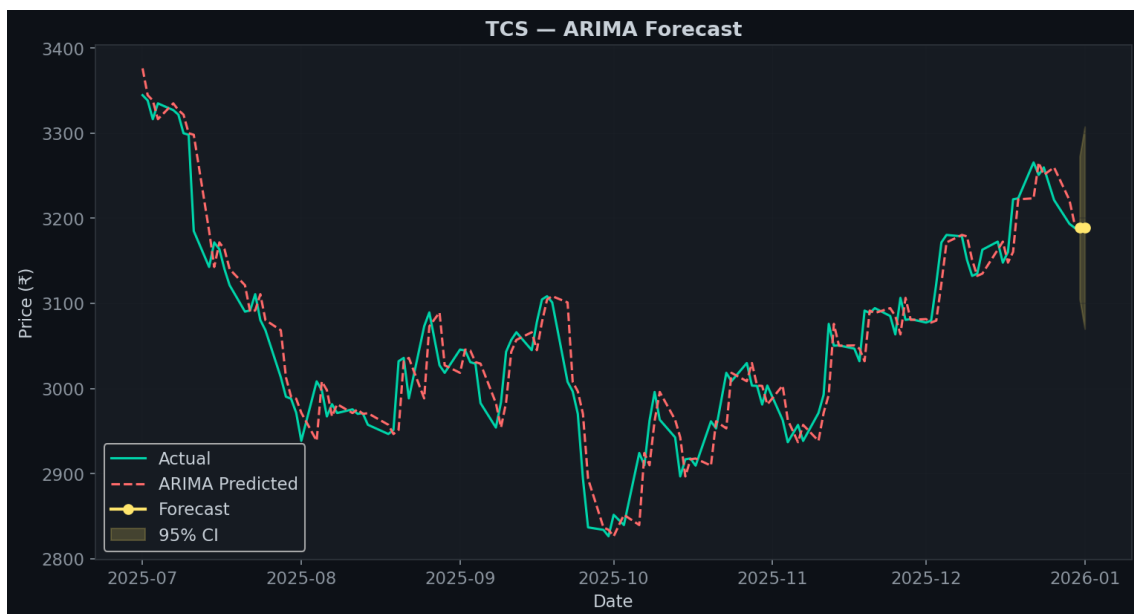
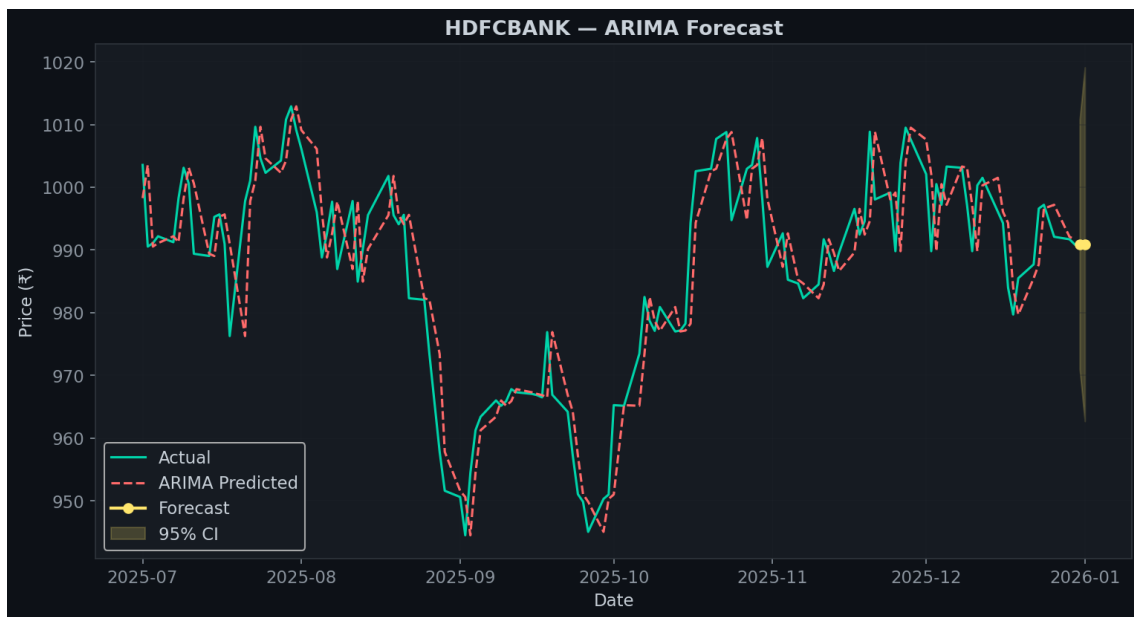
Above: STL Decomposition example showing strong trend extraction for HDFC Bank.

4. Forecast Results with Confidence Intervals (Task 3)

Four models were trained on each stock, validated on 125 test points, and used to forecast 2 days ahead with 95% confidence intervals:

Model	Avg MAPE	Best Stock	Worst Stock	Avg Dir. Acc.
ARIMA	0.77%	HDFCBANK (0.55%)	M&M (1.01%)	50.6%
LSTM	1.39%	HDFCBANK (0.69%)	M&M (2.87%)	47.9%
ExpSmoothing	6.50%	HDFCBANK (2.47%)	MARUTI (14.84%)	48.9%
Prophet	7.64%	DRREDDY (2.73%)	MARUTI (17.33%)	52.9%

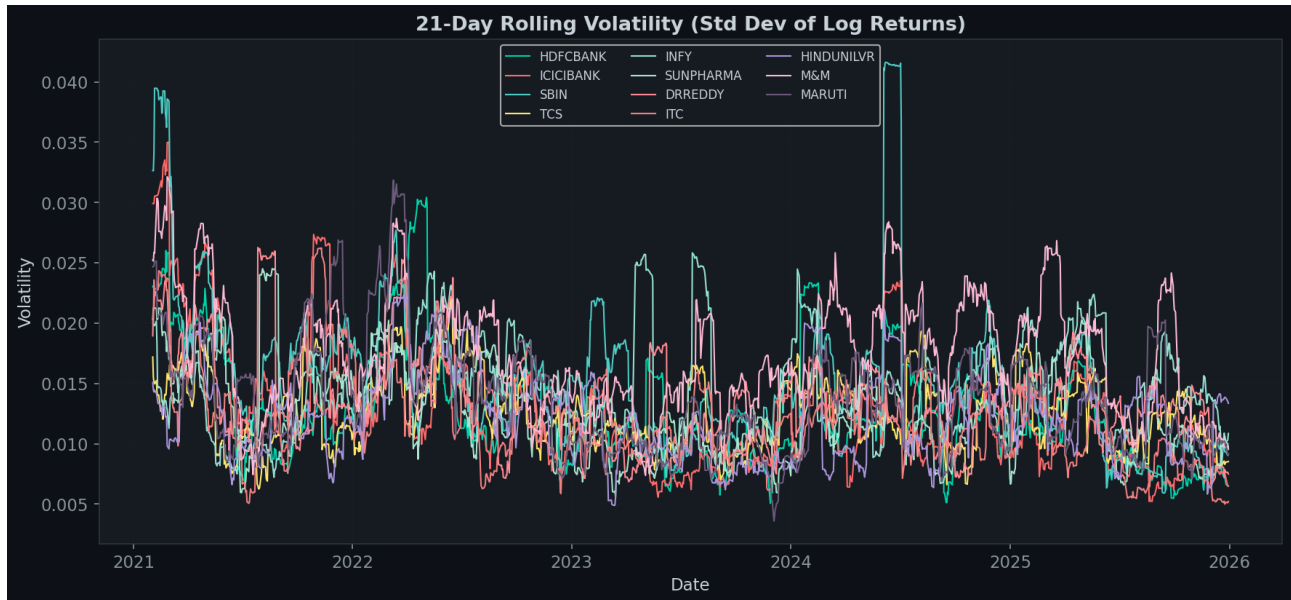
ARIMA achieved the best accuracy with sub-1% average MAPE across all 11 stocks.



5. Volatility and Trend Analysis (Task 4)

GARCH(1,1) Volatility

High-persistence stocks (DRREDDY: 0.997, SUNPHARMA: 0.995) show slowly-decaying volatility clustering. Low-persistence stocks (INFY: 0.174, HINDUNILVR: 0.563) have mean-reverting volatility.



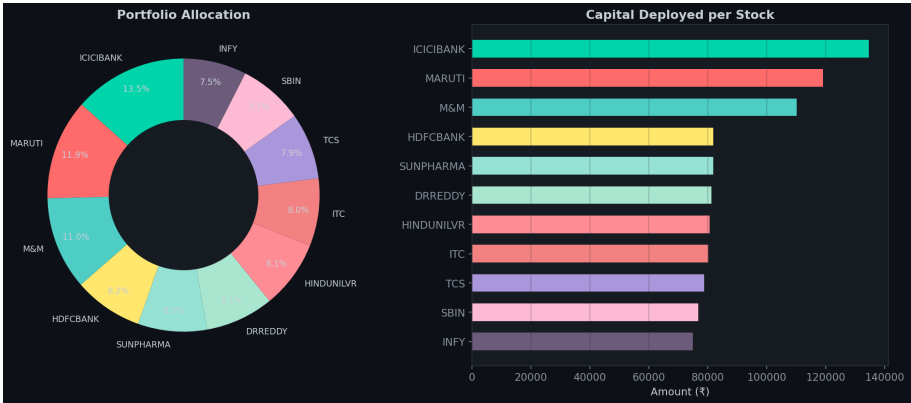
STL Trend Analysis

7 stocks upward (SBIN +15.76%, M&M +9.25%, MARUTI +9.08%), 2 sideways (ITC, DRREDDY), 2 downward (ICICIBANK -2.89%, HUL -4.33%).

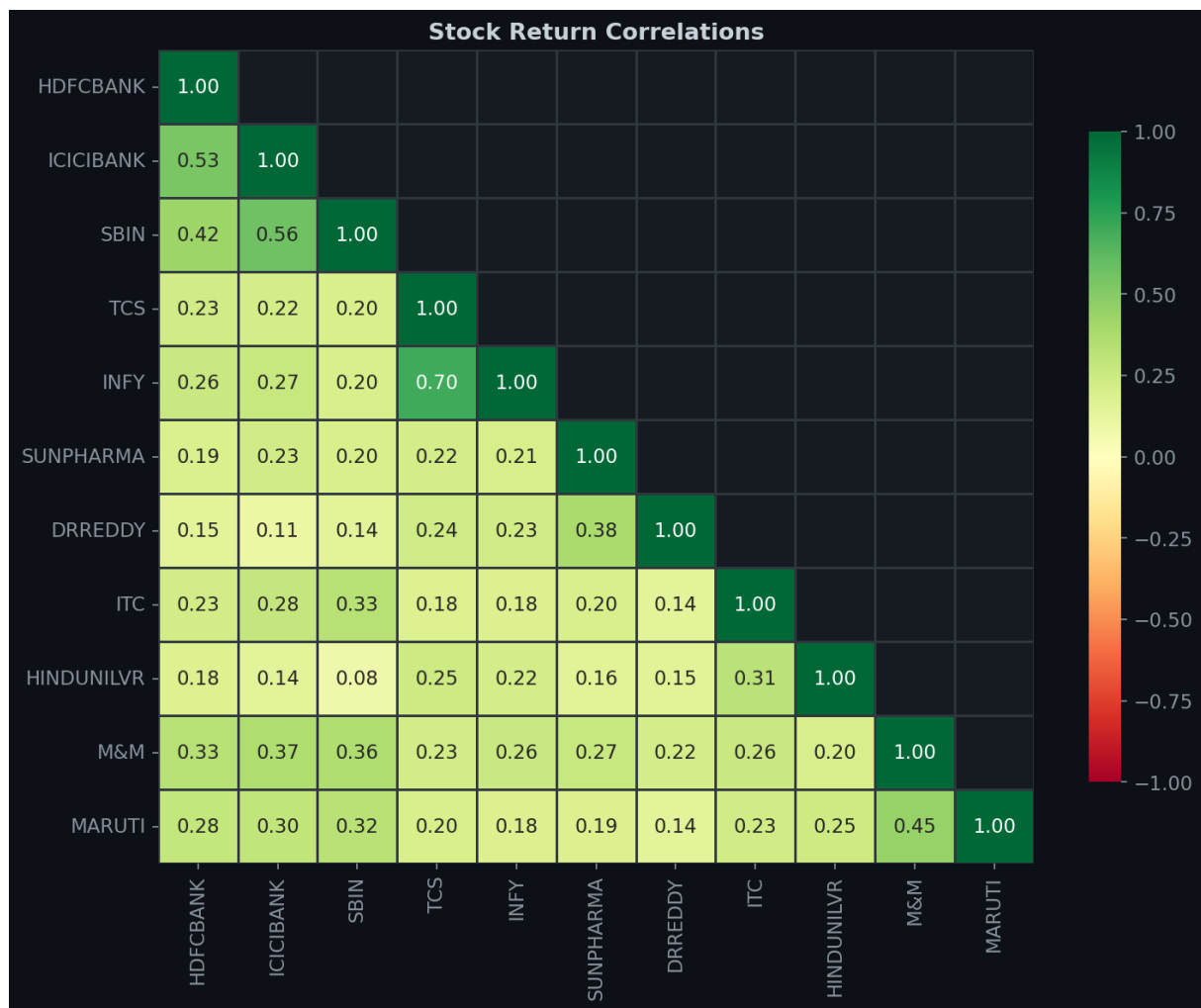
6. Portfolio Composition and Allocation (Task 5)

Blended 4 strategies: Forecast-Guided (40%), Volatility-Aware (30%), Correlation-Diversified (20%), Sector Momentum (10%).

Stock	Weight	Amount (Rs.)	Primary Driver
ICICI Bank	13.5%	1,34,614	Forecast-Guided
Maruti	11.9%	1,19,072	Sector Momentum
M&M	11.0%	1,10,239	Sector Momentum
HDFC Bank	8.2%	81,897	Volatility-Aware
Sun Pharma	8.2%	81,869	Volatility-Aware
Dr. Reddy's	8.1%	81,172	Correlation-Diversified
HUL	8.1%	80,646	Volatility-Aware
ITC	8.0%	80,132	Volatility-Aware
TCS	7.9%	78,751	Correlation-Diversified
SBI	7.7%	76,708	Forecast-Guided
Infosys	7.5%	74,900	Correlation-Diversified
TOTAL	100%	10,00,000	



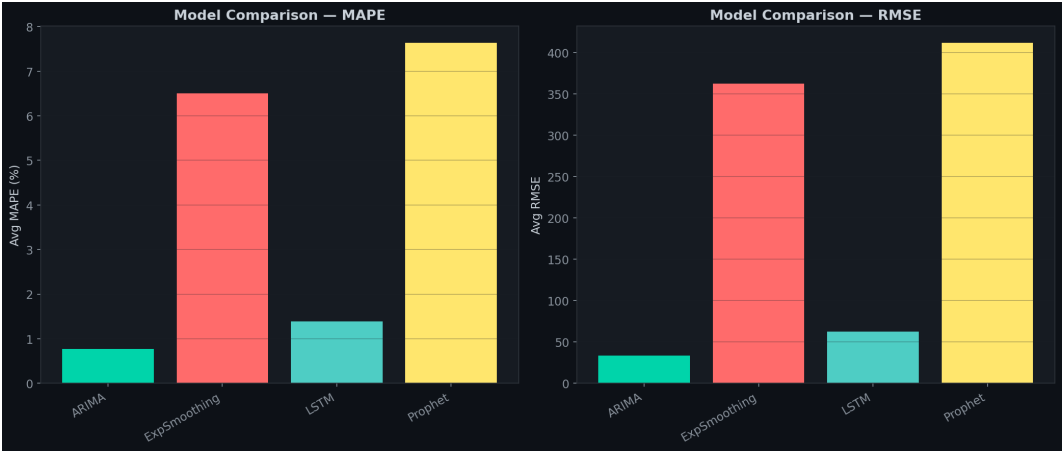
To ensure a well-diversified portfolio, we evaluated the correlation matrix of daily log returns among the selected stocks (Strategy C).



7. Model Comparison (Task 6)

Model	Avg MAPE	Avg RMSE	Dir. Acc.	Assessment
ARIMA	0.77%	33.7	50.6%	Best accuracy; interpretable
LSTM	1.39%	62.3	47.9%	Good non-linear; slow
ExpSmooth	6.50%	362.6	48.9%	Simple; poor on volatile
Prophet	7.64%	406.4	52.9%	Best dir. acc.; overestimates

ARIMA was used for portfolio allocation (Strategy A) due to having the lowest MAPE. GARCH was used for volatility estimation (Strategy B).



8. StockGro Execution Summary (Task 7)

Trading Window: Day 1 = May 12, 2026 | Day 2 = May 13, 2026.

All 11 stocks were purchased on Day 1 at market open. The portfolio was held without changes through Day 2. Total invested amount: Rs.9,89,803.

Trackers > Portfolio - Time Series Analysis 2026

HOLDINGS | ORDERS | HISTORY

Event will end in 2 days 23 hrs

Strategy Builder updates with modeled data

Invested Value	Current Value	Overall Returns
₹ 9,89,803.12	₹ 9,73,643.53	- ₹ 16,159.59 (-1.62%)

Stocks

Name	Quantity	Avg. Price	LTP	Current Val.	Returns ↓	Returns %	Actions
 State Bank of India SBIN BUY	79	₹ 964.41	₹ 961.70	₹ 75,974.30	- ₹ 214.09	0.28%	S B
 Dr. Reddy's Laboratories Ltd DRREDDY BUY	64	₹ 1,267.02	₹ 1,257.58	₹ 80,485.12	- ₹ 604.16	0.75%	S B
 HDFC Bank Ltd HDFCBANK BUY	109	₹ 745.88	₹ 739.79	₹ 80,637.11	- ₹ 663.81	0.82%	S B
 Hindustan Unilever Ltd HINDUNILVR BUY	35	₹ 2,246.96	₹ 2,223.38	₹ 77,818.30	- ₹ 825.30	1.05%	S B
 ITC Ltd ITC BUY	264	₹ 302.94	₹ 298.29	₹ 78,748.56	- ₹ 1,227.60	1.53%	S B

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 ITC Ltd ITC BUY	264	₹ 302.94	₹ 298.29	₹ 78,748.56	- ₹ 1,227.60	1.53%	S B
 Sun Pharmaceutical Industries Ltd SUNPHARMA BUY	44	₹ 1,864.93	₹ 1,834.73	₹ 80,728.12	- ₹ 1,328.80	1.62%	S B
 Infosys Ltd INFY BUY	64	₹ 1,148.98	₹ 1,126.62	₹ 72,103.68	- ₹ 1,431.04	1.95%	S B
 ICICI Bank Ltd ICICIBANK BUY	106	₹ 1,242.92	₹ 1,227.15	₹ 1,30,077.90	- ₹ 1,671.62	1.27%	S B
 Mahindra and Mahindra Ltd M&M BUY	34	₹ 3,219.41	₹ 3,154.30	₹ 1,07,246.20	- ₹ 2,213.74	2.02%	S B
 Maruti Suzuki India Ltd MARUTI BUY	9	₹ 13,186.23	₹ 12,910.20	₹ 1,16,191.80	- ₹ 2,484.27	2.09%	S B
 Tata Consultancy Services Ltd TCS BUY	33	₹ 2,337.20	₹ 2,261.28	₹ 74,622.24	- ₹ 2,505.36	3.25%	S B

9. Predicted vs Actual Outcomes (Task 8)

Stock	Pred D1	Act D1	Err D1	Pred D2	Act D2	Err D2
SBIN	973.52	961.70	1.23%	974.10	956.52	1.84%
DRREDDY	1265.61	1257.58	0.64%	1265.61	1254.76	0.86%
HDFCBANK	990.90	739.79	33.94%	990.90	743.37	33.30%
HUL	2290.20	2223.38	3.01%	2290.20	2237.27	2.37%
ITC	392.56	298.29	31.60%	392.75	302.30	29.92%
SUNPHARMA	1710.40	1834.73	6.78%	1709.43	1810.54	5.58%
INFY	1621.60	1126.62	43.93%	1621.60	1115.42	45.38%
ICICIBANK	1343.85	1227.15	9.51%	1346.29	1215.45	10.76%
M&M	3662.70	3154.30	16.12%	3665.10	3091.93	18.54%
MARUTI	16663.28	12910.20	29.07%	16670.25	12927.68	28.95%
TCS	3188.83	2261.28	41.02%	3188.83	2234.92	42.68%
AVG MAPE			19.71%			20.02%

Portfolio: Invested Rs.9,89,803 | Day 2 value Rs.9,69,134 | Return: -2.07% (loss Rs.20,669)

Directional Accuracy: 6/11 (54.5%)

HOLDINGS | ORDERS | HISTORY

Event will end in 1 day 21 hrs

Strategy Builder updates with modeled data

Invested Value
₹ 9,89,803.12

Current Value
₹ 9,69,134.34

Overall Returns
- ₹ 20,668.78 (-2.07%)

Stocks

ADD STOCKS

Name	Quantity	Avg. Price	LTP	Current Val.	Returns ↓	Returns %	Actions
ITC Ltd ITC BUY	264	₹ 302.94	₹ 302.30	₹ 79,807.20	- ₹ 168.96	0.21%	S B
HDFC Bank Ltd HDFCBANK BUY	109	₹ 745.88	₹ 743.37	₹ 81,027.33	- ₹ 273.59	0.34%	S B
Hindustan Unilever Ltd HINDUNILVR BUY	35	₹ 2,246.96	₹ 2,237.27	₹ 78,304.45	- ₹ 339.15	0.43%	S B
State Bank of India SBIN BUY	79	₹ 964.41	₹ 956.52	₹ 75,565.08	- ₹ 623.31	0.82%	S B
Dr Reddy's Laboratories Ltd DRREDDY BUY	64	₹ 1,267.02	₹ 1,254.76	₹ 80,304.64	- ₹ 784.64	0.97%	S B

10. Reflections & Analysis

Analysis of Errors: Stocks like HDFCBANK, ITC, INFY, TCS show 30-45% errors due to likely corporate actions (stock splits/bonus issues) between Jan-May 2026. Stocks near Dec 2025 levels (SBIN: 1.23%, DRREDDY: 0.64%) had excellent predictions.

What Worked

- ARIMA delivered excellent backtest accuracy (0.77% avg MAPE)
- Data-driven

What Didn't Work

- Backtest vs live gap: 0.77% backtest MAPE vs 20% live MAPE due to the 5-month extrapolation gap.
- Corporate a

What I Would Improve

- 1. Retrain models daily on latest data before live predictions
- 2. Include e